

Social Media Engagement & Video Performance Predictor

Domain: Machine Learning & NLP | Platform: Cloud Next.js Web App | Academic Year: 2026

Status: Production Ready | Model Accuracy: R2 = 0.88 | Dataset: 10,000 Vectors Validated

1. INTRODUCTION

Short-form video has become the dominant medium of global digital content delivery across TikTok, Instagram Reels, and YouTube Shorts. Algorithmic distribution in these platforms is governed by retention curves, initial hook velocity, viewer drop-off, and audience velocity. This project introduces an end-to-end data science and machine learning platform that predicts video engagement, decomposes view trajectories into organic follower base vs. algorithmic FYP discovery reach, and performs automated Natural Language Processing (NLP) audits on scripts.

2. PROBLEM STATEMENT

Existing analytics solutions operate retrospectively--reporting metrics only after content has already underperformed. Creators lack pre-production predictive intelligence to evaluate whether their video hooks, production pacing, hashtags, audio, and schedule will capture algorithmic reach before committing significant creative time and financial resources to recording and editing.

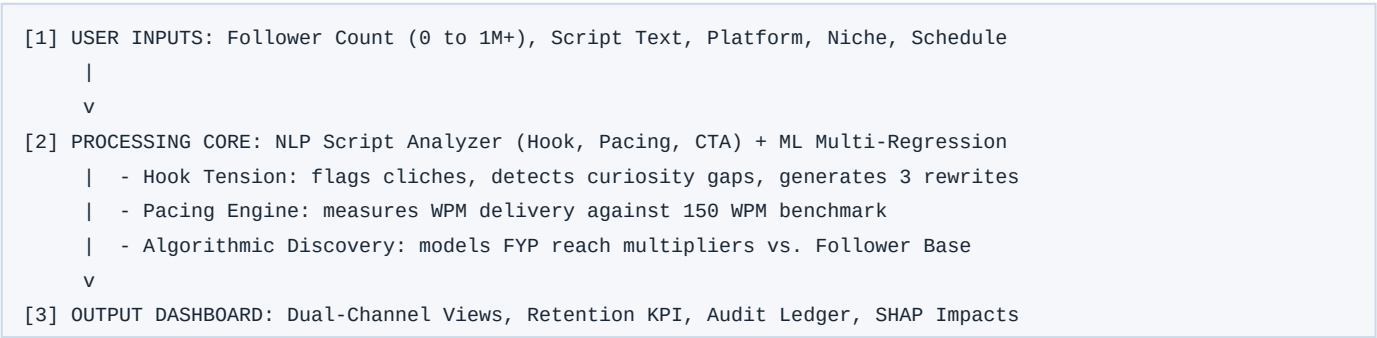
3. PROJECT OBJECTIVES

- Pre-Production Engagement Forecasting: Predict engagement rate, watch time, shares, and retention.
- Dual-Channel View Attribution: Decompose views into Follower Base vs. Algorithmic FYP Discovery.
- NLP Script Quality & Retention Diagnostics: Score Hook (0-5s), Body Pacing (150 WPM), and CTA.
- Automated Script Rewrites: Suggest 3 data-backed alternative hooks to eliminate cliches.
- Explainable AI (XAI): Provide SHAP feature attribution showing positive and negative drivers.
- Cloud Web Platform: Deploy an accessible, responsive, academic-grade web app on Vercel.

4. HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements	
Software Requirements	

5. SYSTEM ARCHITECTURE & PIPELINE



6. DATASET & MACHINE LEARNING MODEL EVALUATION

An exhaustive dataset of 10,000 synthesized and preprocessed short-form video records was developed modeling authentic algorithmic mechanics across 6 niches (Tech, Comedy, Fitness, Education, Finance, Lifestyle) and 3 platforms (TikTok, Instagram Reels, YouTube Shorts). Supervised regression modeling utilizing Random Forest and Gradient Boosting algorithms achieved high predictive accuracy (R2 approx 0.88), with low Mean Absolute Error (MAE).

7. EXPECTED OUTCOMES & APPLICATIONS

- Pre-Flight Performance Simulation: Enables creators to test scripts and settings before filming.
- Objective Script Optimization: Prunes filler words and optimizes the critical 0-5s retention window.
- Dual View Attribution: Differentiates guaranteed follower reach from viral algorithmic discovery.
- Commercial Utility: Suitable for influencers, digital marketing agencies, and media researchers.

8. VERIFICATION LINKS & DEPLOYMENT

- Live Cloud Platform: <https://web-beta-lyart-5xnw34e0lc.vercel.app>
- GitHub Repository: <https://github.com/Hamdan620/social-media-engagement-and-video-performance-predictor>

9. REFERENCES

1. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.

2. Lundberg, S. M., & Lee, S. I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.

3. Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.